

Patrick Dyer

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EDUCATION

University of Ottawa

Ottawa, ON

Bachelor of Applied Science, Software Engineering (CO-OP)

SKILLS & INTERESTS

Skills: Python, Java, Flask, HTML, CSS (Basic), TypeScript, Windows, VS Code, Github, SQL, Android Studio, Communication, Teamwork, Technical writing

Interests: Software development, API deployment, Collaborative projects, Tutoring

Coursework: Data Structures, Computer Architecture, Algorithms Analysis, Software Methodology, Artificial Intelligence

PROJECT EXPERIENCE

Personal Projects

Ottawa, ON

Developer, Designer

Jul 2025 - Present

- UOScheduler (TypeScript): Generates and validates course schedules with conflict detection, and filtering for user choice
- Applied back-end logic and front-end integration for a user-friendly scheduling tool
- Developed an intuitive web application that automates university schedule generation, reducing planning time by 40%.
- Optimized the backend algorithms to efficiently process complex student preferences, decreasing server response times by 35% and supporting scalability for future enrollment growth.

Developer

Jan 2025 - Apr 2025

- Parking Lot Simulator and Optimizer (Java): Applied 6+ Object Oriented Programming principles
- Implemented binary search tree algorithms to optimize parking lot simulation, reducing the queue wait time by 25%
- Gained experience using 5+ data structures (queues, stacks, lists) and exception handling

WORK EXPERIENCE

Transportation Safety Board of Canada

Ottawa, ON

Data Solutions and Development Intern

Jan 2026 - Apr 2026

- Engineered complex SQL queries to extract and manipulate large-scale datasets, powering dynamic data visualization dashboards to track critical transportation safety metrics.
- Developed custom VBA macros to automate complex Excel workflows, building tools for multiplicity flagging and dynamic index generation to efficiently manage hierarchical aviation records.
- Streamlined data reporting processes by bridging raw database information with accessible formats, ensuring high data integrity and reducing manual processing time.

The Dugout

Belleville, ON

Server

Mar 2023 - Aug 2025

- Delivered customer service in a fast-paced environment, ensuring accuracy and attention to detail
- Coordinated with team members to manage high-volume shifts effectively
- Developed interpersonal and communication skills by engaging with diverse clientele

LEADERSHIP EXPERIENCE

UOttaHack7

Ottawa, ON

Participant

Jan 2025 - Jan 2025

- Attended API workshops to experience how to deploy your own API
- Gained hands-on experience in a group setting, developing ideas for a client-based project